City Hospital: Data Preprocessing, ABT Engineering & Exploratory Insights

Prepared by: Data Preprocessing (DP) Team | **Observation Window:** January – June 2024 | **Reference Date:** July 1, 2024

1. Executive Summary of Data Quality Issues & Cleaning Methodology

The raw dataset (*Project_Data.xlsx*) comprises 10,008 consultation records across 500 unique patients and 16 attributes. Prior to aggregation, a systematic data quality audit uncovered six structural categories of anomalies. In compliance with hospital governance, **zero patients were dropped** (all 500 unique Patient IDs are preserved in the final ABT).

• **Exact Duplicate Consultations:** Detected and removed 8 exact duplicate rows, reducing the consultation table from 10,008 to 10,000 unique visits.

• **Conflicting Static Patient Attributes:** Static attributes varied across visits for specific patients (Gender: 5 patients, Marital Status: 4, Profession: 2, Age: 0). Resolved chronologically by sorting by *Visit Date* and retaining each patient's most recent non-null record, followed by pediatric consistency rules (patients under 18 standardized to Student / Single).

• **Impossible Age & Rating Values:** Flagged 103 records with biologically impossible ages (118, 138, 142, 158, 195 years caused by leading '1' digit typos) as NaN prior to imputation. Capped 719 out-of-scale Satisfaction Level entries (recorded as 6) to the valid maximum of 5.

• **Continuous Outliers (1.5x IQR Method):** Winsorized 10 extreme Consultation Duration records (up to 600 min) at the upper IQR fence of 173.5 min. Winsorized 7 Consultation Price anomalies (including 7 decimal-shift typos between £25,300 and £39,999) using department-specific 1.5x IQR fences. Flagged 127 anomalous adult income records (> £75,534 up to £11.97M) as NaN for predictive imputation while preserving legitimate £0 incomes for Students.

• **Missing Value Treatment (KNN Imputation & Domain Logic):** Treated structural missingness in Family History (1,993 rows / 99 patients) as *'None'* and Insurance Provider (92 patients with £0 coverage) as *'Uninsured'*. Applied scikit-learn's distance-weighted **KNNImputer (k=5)** across encoded demographic and financial features to impute missing values in Age (56 rows), Education Level (29 rows), Approximate Annual Income (154 rows), and Insurance Coverage (50 rows). Finally, capped 8 rows where Insurance Coverage exceeded Consultation Price.

2. Analytic Base Table (ABT) Construction & Data Dictionary

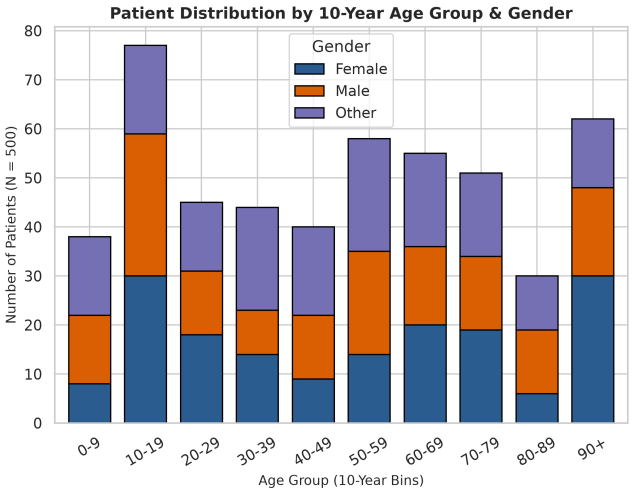
The cleaned visit records were aggregated by *Patient ID* into **ABT_city_hospital.csv** (500 rows x 37 engineered features), capturing static demographics, consultation behaviour, weekend vs. weekday habits, financial exposure, satisfaction pain points, and recency metrics relative to **July 1, 2024**.

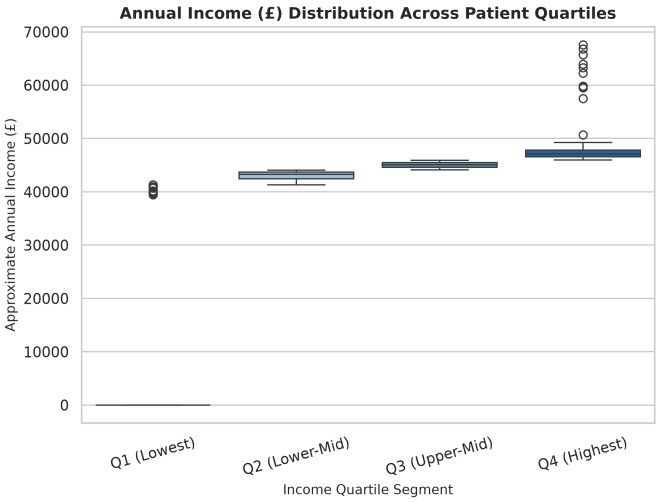
Feature Name	Category	Definition & Calculation Logic
Patient ID	Static / ID	Unique identifier for each patient (1 row per patient, N=500).
Age	Static Demographic	Patient age in years (most recent valid record, KNN-imputed if missing).
Gender	Static Demographic	Patient gender (Female, Male, Other); standardized to most recent visit.
City_of_Residence	Static Demographic	Patient city of residence across 8 UK/Irish cities.
Profession	Static Socio-Economic	Patient occupation (standardized to most recent visit; <18 set to Student).
Insurance_Provider	Static Financial	Provider A, B, C, D, or 'Uninsured' for patients with no policy.
Family_History	Static Clinical	Recorded family medical history (Cancer, Diabetes, Heart Disease, Hypertension, None).
Education_Level	Static Socio-Economic	Highest educational attainment (Pre-school to PhD; KNN-imputed).
Marital_Status	Static Demographic	Marital status (Single, Married, Divorced, Widowed; standardized).
Approx_Annual_Income	Static Socio-Economic	Mean cleaned annual income (£) across patient consultations.
Age_Group_10yr	Static Derived	10-year age bracket (0-9, 10-19, ..., 90+) for demographic segmentation.
Income_Quartile	Static Derived	Patient income quartile (Q1 Lowest to Q4 Highest).

Feature Name	Category	Definition & Calculation Logic
Total_Visits	Visit Behaviour	Total number of hospital consultations between Jan 1 and Jun 30, 2024.
Distinct_Departments	Visit Behaviour	Number of unique hospital departments visited by the patient.
Most_Frequent_Department	Visit Behaviour	Modal (most frequently visited) department for the patient.
Avg_Consultation_Duration	Visit Behaviour	Mean consultation duration in minutes (post-IQR Winsorization).
Total_Consultation_Duration	Visit Behaviour	Cumulative consultation duration in minutes across all visits.
Weekend_Visits	Visit Behaviour	Count of consultations occurring on Saturday or Sunday.
Weekday_Visits	Visit Behaviour	Count of consultations occurring Monday through Friday.
Weekend_to_Weekday_Ratio	Visit Behaviour	Ratio of weekend visits to weekday visits (Weekend / max(1, Weekday)).
Weekend_Visit_Share	Visit Behaviour	Proportion of total visits occurring on weekends.
Total_Consultation_Price	Financial	Cumulative gross consultation cost (£) across all visits.
Avg_Consultation_Price	Financial	Mean gross price (£) per consultation.
Total_Insurance_Coverage	Financial	Cumulative amount (£) covered and paid by the insurance provider.
Avg_Insurance_Coverage	Financial	Mean insurance coverage amount (£) per consultation.
Total_Out_of_Pocket	Financial	Total out-of-pocket expenditure (£) paid by the patient (Price - Coverage).
Avg_Out_of_Pocket	Financial	Mean out-of-pocket cost (£) per consultation.
Out_of_Pocket_Ratio	Financial	Share of total consultation price paid out-of-pocket by the patient.
Avg_Satisfaction_Rating	Satisfaction	Mean consultation satisfaction score (1 = lowest, 5 = highest).
Min_Satisfaction_Rating	Satisfaction	Minimum satisfaction rating recorded across any visit.
Max_Satisfaction_Rating	Satisfaction	Maximum satisfaction rating recorded across any visit.
Low_Rating_Count	Satisfaction	Total count of dissatisfied visits with rating <= 2.
Share_Low_Ratings	Satisfaction	Proportion of patient visits rated <= 2 (dissatisfaction rate).
Days_Since_First_Visit	Recency & Activity	Days elapsed from the patient's first visit to July 1, 2024.
Days_Since_Last_Visit	Recency & Activity	Days elapsed from the patient's most recent visit to July 1, 2024.
Active_Span_Days	Recency & Activity	Number of days between the patient's first and last recorded visit.
Visits_Per_Month	Recency & Activity	Average monthly consultation frequency over the 6-month period (Total_Visits / 6).

3. Exploratory Visualizations & Key Analytical Findings

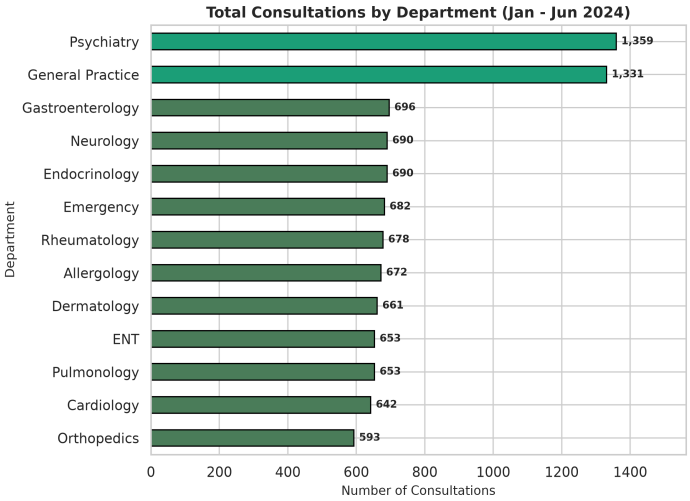
3.1 Patient Demographics: Age Bins & Income Quartiles

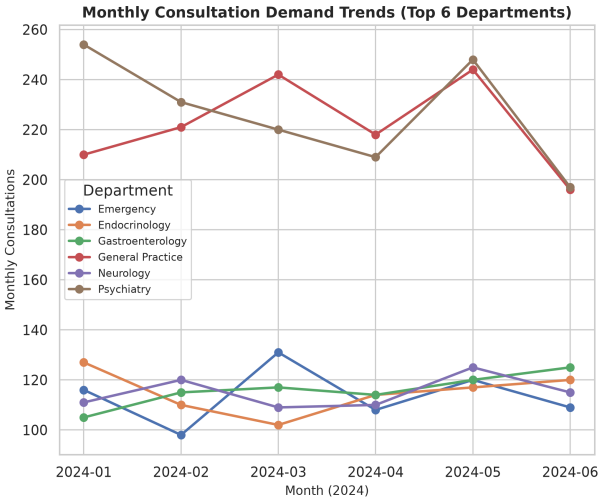




City Hospital serves a broad demographic spanning infants to seniors (mean age: 49.1 yrs), with balanced gender representation across 10-year age brackets. Income quartiles reflect the structural split between dependent/student cohorts in Q1 (£0 income) and working/retired adults earning £30k–£60k across Q2–Q4.

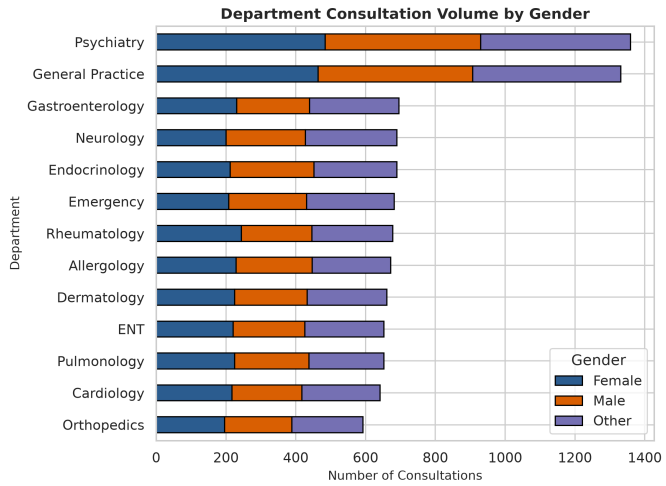
3.2 Department Demand & Monthly Trends (January – June 2024)

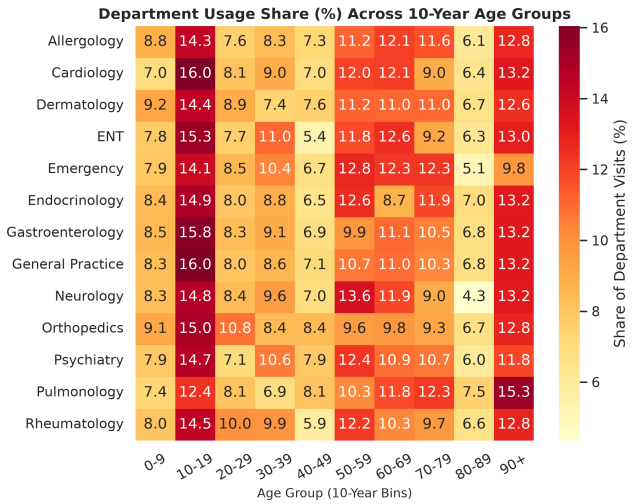




Psychiatry (1,359 visits, 13.6%) and **General Practice** (1,331 visits, 13.3%) are the hospital's two busiest departments, each handling roughly double the volume of any other specialty (~600–700 visits each). Monthly demand remains steady from January through June with a slight dip in February due to fewer calendar days.

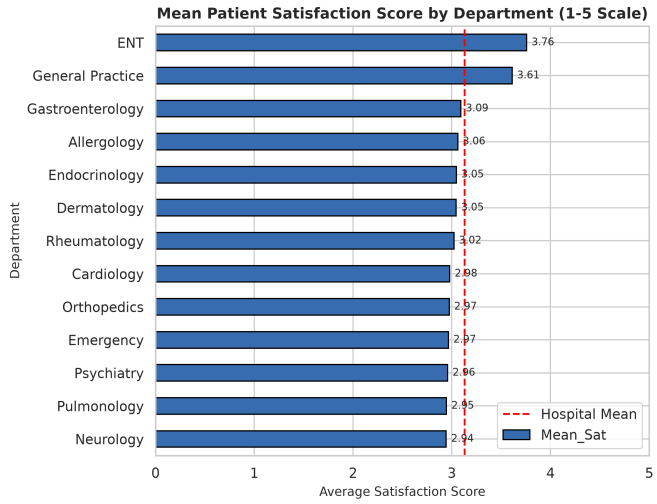
3.3 Department Utilization by Gender & 10-Year Age Group

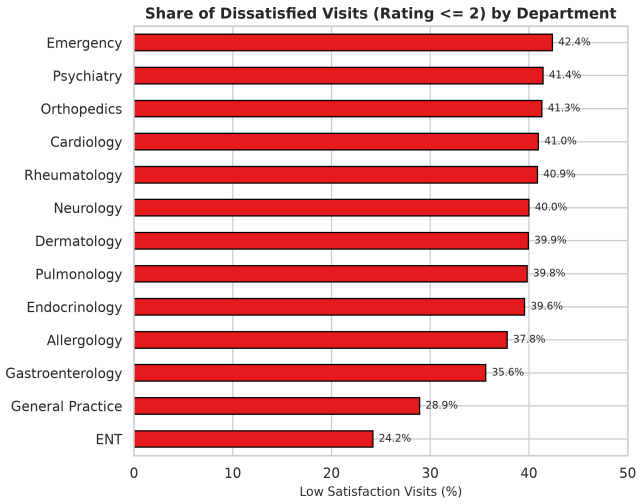




Consultation volume across all 13 departments is evenly utilized across Female, Male, and Other genders, and distributed proportionally across pediatric, working-age, and senior age brackets. Multi-department utilization is high: patients visit an average of 9.9 distinct departments over the 6-month period (20.0 total visits/patient).

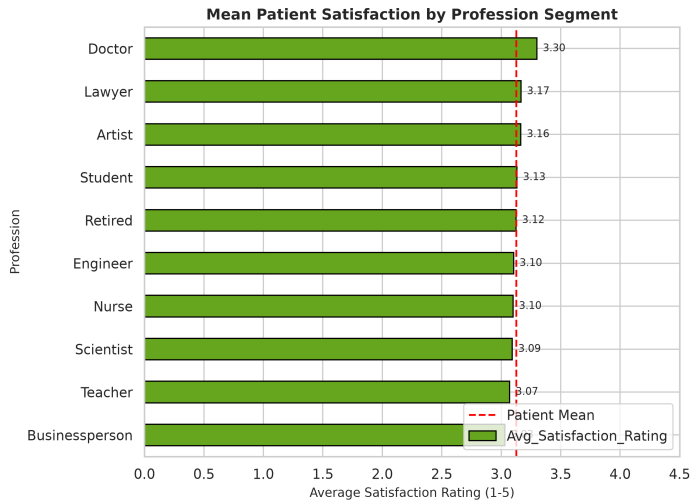
3.4 Patient Satisfaction & Low-Rating Pain Points by Department

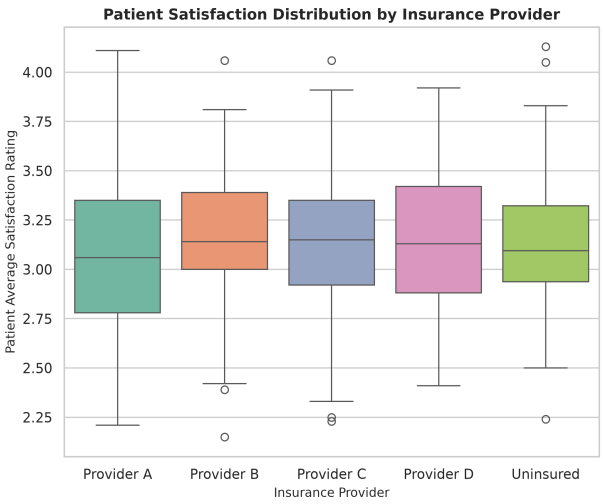




Hospital-wide satisfaction averages 3.13/5.00, with 37.5% of visits rated <= 2. Across departments, **Neurology** reports the lowest mean satisfaction (2.94), whereas **ENT** achieves the highest rating (3.76). Addressing dissatisfaction in high-volume departments (Psychiatry & General Practice) offers the largest overall quality impact.

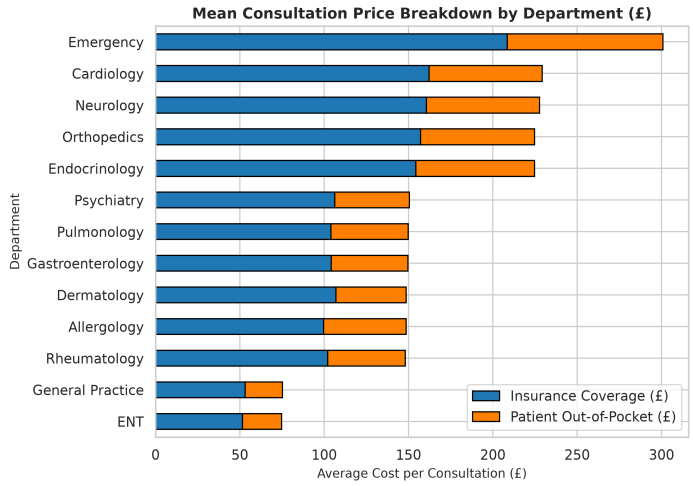
3.5 Patient Satisfaction Across Socio-Economic & Insurance Segments

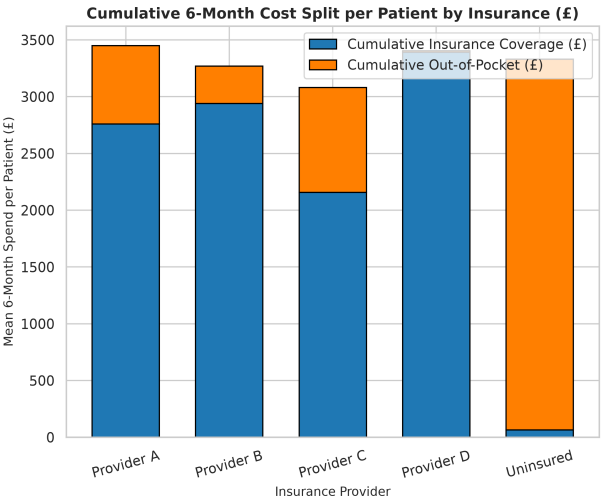




Patient-level satisfaction varies across occupation and coverage tiers. Because every patient visits multiple departments over 6 months, dissatisfaction is driven primarily by service experience at the visit level rather than demographic bias, though uninsured and lower-coverage patients face much steeper financial friction.

3.6 Financial Breakdown: Insurance Coverage vs. Out-of-Pocket Burden





Across all 10,000 consultations (gross billings: £1,651,924), insurance covers **69.8%** (£1,152,924) while patients pay **30.2%** (£499,001) out-of-pocket. Emergency (£300 avg), Cardiology, Neurology, Endocrinology, and Orthopedics (£225 avg) are the highest-cost departments, while ENT and General Practice (£75 avg) are the most affordable. Uninsured patients bear nearly 100% of their consultation costs out-of-pocket (~£3,100 cumulative over 6 months).

4. Executive Answers to Management Questions

Q1: Which departments are the busiest, and how does demand change from January to June?

Answer: *Psychiatry* (1,359 visits) and *General Practice* (1,331 visits) are by far the busiest departments, accounting for 26.9% of all hospital consultations. The remaining 11 specialist departments handle between 593 (Orthopedics) and 696 (Gastroenterology) visits each. Demand is structurally consistent from January through June (~1,650–1,720 total visits/month, with a minor calendar dip in February), and 27.8% of consultations occur on weekends.

Q2: Which patient groups use which departments most?

Answer: Service utilization is broad-based: patients average 20.0 visits across 9.9 distinct departments over 6 months. Retirees (36.0% of visits) and Students (21.0% of visits) represent the largest occupational user groups. Both groups heavily utilize Psychiatry and General Practice as primary care hubs before branching into specialist departments.

Q3: Which departments and patient groups report the lowest satisfaction?

Answer: Hospital-wide, 37.6% of visits receive a low rating (<= 2). *Neurology* records the lowest departmental mean rating (2.94/5.00), closely followed by Pulmonology and Cardiology. Because high-volume departments (Psychiatry and General Practice) also see ~37% low ratings, they generate the highest absolute volume of dissatisfied patient encounters.

Q4: How much of the consultation cost is covered by insurance vs. paid out-of-pocket?

Answer: Across all consultations, mean gross price is £165.19, of which insurance covers £115.29 (**69.8%**) and patients pay £49.90 (**30.2%**) out-of-pocket. Insured patients (Providers A–D) have ~83% of their costs covered, whereas the 92 Uninsured patients pay nearly 100% out-of-pocket.

5. Strategic Recommendations for the Next Analytics & Modeling Team

- **Candidate Clustering / Segmentation Frameworks:** Use the ABT to run K-Means or Gaussian Mixture clustering along three orthogonal axes: (1) *Financial Vulnerability & Care Burden* (Total_Out_of_Pocket, Out_of_Pocket_Ratio, Uninsured status) to identify patients needing financial counseling; (2) *Multi-Department Chronic Utilizers* (Total_Visits, Distinct_Departments, Total_Consultation_Duration) for integrated care pathways; and (3) *Weekend-Dependent Working Professionals* (Weekend_Visit_Share, Profession) for extended-hours scheduling.

• **Supervised Predictive Modeling Targets:** (1) *Patient Dissatisfaction Risk Model*: Predict whether a patient will have a high Share_Low_Ratings (> 0.40) to trigger proactive patient-experience outreach; (2) *High-Cost / High-Utilization Forecasting*: Predict *Total_Consultation_Price* or *Visits_Per_Month* to optimize departmental staffing and resource allocation.

• **Data Capture & Engineering Improvements:** Enforce front-end EHR validation rules to block ages > 110, satisfaction scores > 5, consultation prices > £500, and coverage exceeding price at the point of entry.